Plant Soil
https://doi.org/10.1007/s11104-025-08048-4

REVIEW ARTICLE

Prospecting silicon-based agroecological agroforestry systems for counteracting climate change

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Received: 29 April 2025 / Accepted: 22 October 2025
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Abstract

Background Most research on silicon (Si) has focused on high or intermediate Si-accumulating species, leaving a knowledge gap on the role of Si in low Si-accumulating plants, fruit trees and other woody species. However, Si can provide ecological and agricultural benefits across a broad range of families. Agroecological (AE) and agroforestry (AF) systems, particularly in Mediterranean climates, are of great interest for addressing the challenges posed by climate change and land degradation and offer opportunities to integrate knowledge on Si to enhance agroecosystem health.

Scope This review provides insights into i) Si accumulation in fruit trees, woody forestry species, timber species, and shrubs and its benefits in fruit trees, ii) Si cycling in the plant-soil system, iii) and discusses the potential for Si-mediated improvements in AE and AF systems.

Conclusion The review underscores benefits of Si application on fruit tree crops, high Si-accumulation and phytolith formation in woody species, while discussing how knowledge on Si cycling in forest ecosystems and rhizosphere processes can be applied in AE and AF systems. Silicon accumulation by deep-rooted woody species may lead to increased availability of Si and other elements by mineral weathering, carbon (C) sequestration, and Si-recycling by litter or root turnover, which could be exploited, together with soil management, to improve Si nutrition in intercropped species. The adoption of agronomic strategies, including Si-efficient rootstocks, appears relevant for improving agroecosystem resilience, especially in climate vulnerable ecosystems like the Mediterranean region.

Keywords Phytolith · Fruit trees · Woody species · Intercropping · Rhizosphere

Responsible Editor: Martin J. Hodson.

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Introduction

Extreme conditions, such as heat waves, drought, flooding, fires, among others, intensified by climate change, threaten plant development and agricultural productive capacity. Addressing these challenges requires understanding plant processes and responses to environmental pressures (Jing et al. 2024), along with the development of sustainable and adaptive strategies.

Silicon (Si) has gained increasing attention for its proven ability to improve plant growth and stress tolerance by influencing physiological and biochemical

Published online: 20 November 2025

processes, with beneficial impacts on agricultural production (Ma 2004; Epstein 2009; Thakral et al. 2024; Etesami 2025). Silicon accumulation varies widely among species, with plants commonly classified based on their Si content (Ma et al. 2001). However, the majority studies have been focused on high (> 1%) or intermediate (0.5–1%) Si-accumulating plant species (Mitani-Ueno and Ma 2021; Thakral et al. 2024), leaving a gap on the benefits and roles of Si in species considered as low accumulators (Katz 2014; Putra et al. 2020).

Increasing evidence indicate agricultural and ecological importance and benefits of Si in a variety of plant families and species such as non-grass taxa (Katz 2014; Hoffmann et al. 2020; Katz et al. 2021), nitrogen-fixing herbaceous and woody legumes (Putra et al. 2020, 2024; Gonzalez-Porras et al. 2024), fruit trees (Etesami and Jeong 2020; Muhammad et al. 2022; Rachappanavar et al. 2024) and forest tree species (Bartoli 1983; Cornelis et al. 2010b; Schaller et al. 2018; Kothari et al. 2025).

Fruit trees and other woody species are integral to many agroecosystems (Ramachandran et al. 2017; Lehmann et al. 2020; Barbault et al. 2024), especially in Mediterranean regions where sustainable strategies are increasingly needed due to climate change (Medda et al. 2022). These landscapes have long been shaped by traditional woody-crop systems (Ramachandran et al. 2017), reflecting centuries of integrated land use and whose revalorization now highlights their climate resilience and ecosystem services (Quandt et al. 2023). Despite this, Si dynamics in fruit trees remains understudied and information on their Si-accumulating capacities is limited (Katz 2014; Etesami and Jeong 2020). This knowledge is essential for understanding Si-soil–plant interactions and the potential of these species to contribute to Si cycling in agroecosystems.

Emerging research has highlighted the functional role of deep-rooted woody species in accessing subsoil Si pools, promoting mineral weathering, and increasing Si availability through root-microbe interactions and litter recycling (Putra et al. 2024; Li et al. 2025; Kothari et al. 2025). Notably, the multiple ecological roles of Si in forest ecosystems, Si dynamics and biogeochemistry across different forest ecosystems have been recently reviewed (Kothari et al. 2025). Phytolith accumulation in woody species, and the implications of phytolith-occluded carbon (PhytOC) in carbon (C) sequestration are relevant topics in the context of climate change (Hodson 2019; de Tombeur et al. 2024; Kothari et al. 2025). Phytolith formation has been documented in many woody species, including Mediterranean fruit trees such as *Vitis vinifera*, *Olea europaea*, and *Ficus carica* (Tsartsidou et al. 2007; Hermans et al. 2025), indicating their potential role in Si cycling and C storage.

Therefore, in-depth knowledge of Si accumulating processes in these species and agroecological (AE) and agroforestry (AF) systems present promising avenues to harness Si-mediated processes. Noteworthy, intercropping with Si-accumulating species enhances nutrient availability, pest resistance, and plant performance through rhizosphere interactions, while reducing dependency on external inputs (Ning et al. 2022; Fujii et al. 2023; Putra et al. 2024). These findings have been demonstrated on herbaceous crops, whereas similar information on woody crops is scant. Agroforestry systems, characterized by the presence of woody species together with herbaceous crops and eventually animals, provide multiple benefits including enhanced nutrient cycling, improved soil fertility and C sequestration, reduced erosion, increased water-use efficiency, while also supporting climate change mitigation (Quandt et al. 2023; Isaac et al. 2024; Nungula et al. 2024). The benefits of AF systems compared to monocultures and the shortage of scientific information on Si make them particularly relevant for studying Si-mediated processes. Integrating Si management into Mediterranean agroecosystems can increase crop resilience (Etesami 2025).

To address the above mentioned gaps, this review aims to provide a current overview on: (i) Si accumulation in fruit trees, woody forestry species, timber species, and shrubs and its physiological and agronomic benefits in fruit trees; (ii) Si cycling in the plant-soil system; and (iii) the potential integration of Si-based mechanisms into AE and AF systems, emphasizing their contribution for sustainable agriculture in climate-vulnerable regions like the Mediterranean. These systems offer an opportunity to harness ecological processes, such as Si accumulation, C sequestration, root-driven Si mobilization, and Si recycling through litterfall and root turnover, to enhance resilience. While fruit trees and woody species are not traditionally classified as high Si accumulators, their documented capacity to form phytoliths

and response to Si application highlights their potential functional role in Si dynamics in AE and AF systems. By bringing together agronomic, ecological, and biogeochemical perspectives, this review aims to support the design and optimization of diversified agricultural systems that leverage Si-mediated processes for sustainability and productive capacity.

Silicon in plants

Silicon, the second most abundant element in the Earth's crust (Wedepohl 1995), silicic acid ($Si(OH)_4$) is the only plant-available form of Si (Ma et al. 2001), and it's taken up by plants at pH below 9 (Ma and Yamaji 2006).

Research on Si uptake has advanced significantly following the identification in rice (Ma et al. 2006; Yamaji and Ma 2009) of Lsi1, transporter that modulates active Si uptake, homologs have been characterized in several species (Mitani-Ueno and Ma 2021) including woody tree species such as date palm (*Phoenix dactylifera* L.) (Bokor et al. 2019), poplar (*Populus trichocarpa* Torr. & A.Gray ex Hook.) (Deshmukh et al. 2015), grapevine (*Vitis vinifera* L.) and hybrids of *Vitis* species (Noronha et al. 2020). Despite advances, more research is still needed to understand active Si uptake, especially in woody species, as it is believed to be of ecological significance in certain ecosystems (Schaller et al. 2018; Nakamura et al. 2023; Klotz et al. 2023).

Silicon accumulation in plant tissues is involved in structural and functional resilience (Guerriero et al. 2016; de Tombeur et al. 2023). Once taken up, Si influences cell wall elasticity, extensibility and plasticity (Ma and Takahashi 1993; Hossain et al. 2002; Hattori et al. 2003). Silicon affects the synthesis of pectin, polyphenols, cellulose, and lignin, boosting rigidity and flexibility (Sheng and Chen 2020; Zexer et al. 2023). The deposition of SiO_2 is promoted by lignin (Guerriero et al. 2016), and may contribute to wood resilience, essential for tree stability and longevity (Loehle 1988).

A key mechanism of Si deposition in plants is the formation of phytoliths which are silicified structures produced in higher plant tissue (Piperno 1988; Sangster et al. 2001; Schaller et al. 2021). They consist of of $SiO_2 \cdot nH_2O$, organic matter, and various elements, and have been found in almost all plant organs, in sizes ranging from 100 nm up to 1 mm (Schaller et al. 2021). Phytoliths are embedded in the cell walls, cell lumina, and intercellular or cuticular spaces (Hodson 2019; de Tombeur et al. 2024). Silica deposition occurs across tissues but is most concentrated in leaf epidermal cells (Hodson 2019). Plants absorb dissolved Si as monosilicic acid, which moves with the transpiration stream to sites of high water flux, where it polymerizes and precipitates as phytoliths (Piperno 1988; de Tombeur et al. 2024). Upon tissue senescence and litterfall, these phytoliths return to the soil, contributing to the bioavailable Si pool for subsequent plant uptake (Farmer et al. 2005). Recent findings indicate that silica deposition is not purely passive but involves active control by specific polymers and proteins that concentrate and stabilize silicic acid, enabling its condensation into silica and subsequent deposition (Zexer et al. 2023).

The quantity, composition, morphology, and anatomical localization of phytoliths are variable within and between species and at tissue-level (de Tombeur et al. 2024). Interspecific differences in phytolith chemistry and C content, such as higher C concentrations in wall-embedded phytoliths than in lumen types, have been documented (Song et al. 2014; Yang et al. 2015; de Tombeur et al. 2024). However, C sequestration potential may be underestimated due to limited data on phytoliths in roots, woody tissues, and belowground biomass (Cheng et al. 2023). Existing literature remains skewed toward grasses, croplands, and a few forest types (Cheng et al. 2023), less information is available on fruit trees and woody species, particularly in Mediterranean agroecosystems, including archeological studies using phytolith for botanical identification (Tyree 1994; Albert and Weiner 2001; Tsartsidou et al. 2007; Castilla-Beltrán et al. 2024; Hermans et al. 2025).

Si accumulation, deposition and C-sequestration processes highlights that Si uptake and phytolith formation are not only fundamental to plant structure and stress tolerance but also central to long-term nutrient and C dynamics.

Si accumulation and phytolith occurrence in fruit trees and woody species

Si accumulation

As previously highlighted, Si accumulation in woody species, particularly fruit trees, remains understudied

despite their ecological and agricultural significance. A tailored Web of Science search (last updated on March 2025) using the search statement = ((accumulation or uptake or accumulated or translocation) AND (silicon or silica or 'Si' or silicification or silicate or silicic) AND (tree or trees or woody)), defined to journal articles and conference proceedings, revealed limited data on Si in woody species. The initial meta-analysis search produced 292 results, which after individual article consulting for explicit data on accumulation of Si, the results were narrowed down to only 27, of which just 13 were focused on fruit tree species. An additional tailored search was performed for fruit trees (Table 1). The majority of articles were focused on trees and woody species in natural environments, most likely due to the frequent use of common or scientific names.

The compiled literature reveals wide differences in leaf Si concentrations for some of the same species (Table 1), likely due to factors such as different soils, environmental and experimental conditions (Schaller et al. 2021) and plant age and tissue type and age which can lead to discrepancies in plant classification for Si accumulation (Deshmukh et al. 2020). In addition, research has been focused on fruit tree and woody species in temperate (Gérard et al. 2008; Cornelis et al. 2010b, 2011) or tropical (Schaller et al. 2018; Nakamura et al. 2019; Klotz et al. 2023) environments, leaving Mediterranean climates and regions underrepresented (de Tombeur et al. 2020; Moura et al. 2024).

While research on Si accumulation has predominantly focused on grasses and herbaceous crops, this search highlights the broad range of Si uptake in woody species. Several fruit trees, including *Morus rubra* (1.77%), *Theobroma cacao* (1.29%), and *Vaccinium corymbosum* (6.00%), demonstrate notable Si accumulation, with commonly cultivated species such as *Citrus reticulata*, *Malus pumila*, *Persea americana*, and *Coffea arabica* also exhibiting measurable leaf accumulation. While forest and timber species, particularly *Fagus sylvatica* (up to 11.00%), *Abies alba* (up to 6.25%), *Pseudotsuga menziesii* (4.46%), *Celtis occidentalis* (4.11%), *Picea abies* (0.92%-1.20%) and the sub- and tropical species, *Teijsmanniodendron coriaceum* (12.58%), *Gironniera subaequalis* (6.66%), and *Dipterocarpus cornutus* (6.25%), accumulate substantial amounts of Si, further exemplify the diversity of high-Si accumulators (Table 1). The highest Si concentration values were typically observed in tropical ecosystems (Ishizawa et al. 2019).

In wood, Si varies widely across species and ecosystems, for instance, spruce (*Picea abies* L.) contains three times more Si than birch (*Betula pendula* Roth) or aspen (*Populus tremula* L.) with roughly half of absorbed Si concentrated in woody tissues (Bocharnikova and Matichenkov 2012), indicating the relevance of Si accumulation in woody parts. Analyzing ecosystems reveals that northern hardwood forests have relatively low Si levels (0.009 to 0.023% SiO_2) compared to tropical species where higher concentrations (0.07% to 1.6% SiO_2 dry weight) are found (Eduardo Santana et al. 2013), making Si a major inorganic component in tropical wood and contributing to higher ash content (Feldman 1985). Improved cell wall structural integrity associated with Si concentration could enhance tree resilience to mechanical stress and environmental variability, promoting sustainable forest management practices, highlighting the need to investigate mechanisms of Si accumulation in the wood.

While specific studies on accumulation are few, several studies have explored the efficacy of Si to overcome a diverse range of abiotic and biotic stresses in fruit tree species (Ladon et al. 2021; Muhammad et al. 2022; Rachappanavar et al. 2024) with direct agronomic and physiological implications in fruit crops (Etesami and Jeong 2020). Silicon has been shown to improve growth and yield of fruits trees and woody horticultural crops (Fig. 1), some examples include apple (Karagiannis et al. 2021; Schimmel et al. 2024), avocado (Dann and Le 2017), cherry (Rombolà et al. 2023), chestnut (Carneiro-Carvalho et al. 2019a, b), citrus (Mvondo-She and Marais 2019; ValizadehKaji and Mohammaei 2025), coffee (Parecido et al. 2022), pomegranate (ValizadehKaji et al. 2025), grape (Schabl et al. 2020; Sut et al. 2022), mango (Muhammad et al. 2022), olive (Martos-García et al. 2024), hazelnut (Escobar-Hernández et al. 2025), and others, regardless of plants' ability to accumulate it. Additionally, the importance of Si has been highlighted in the floriculture sector (Kumari et al. 2023) and Si accumulation in flowers has been described as widespread across species (Schoelynck et al. 2023). As flowering has a crucial role in fruit production, these aspects should also be taken into consideration. However, the use of Si in the horticultural sector is emphasized on foliar or fertigation applications, with less importance

Table 1 Leaf Si percentage (%) in woody species described in published literature, with Si concentrations of leaf tissue being reported

Growing Conditions	Woody species	Leaf Si (%[a])	References	Occurrence of phytoliths (Yes/No/Not Available (NA))	References
Agriculturally Important (fruit)		Leaf Si ranges < 0.01–6%; phytolith occurrence in several crops, showing their presence even in species with low leaf Si			
Controlled conditions (Pot)	*Vaccinum corymbosum* L	6.00	(Morikawa and Saigusa 2004)	Yes	(Morikawa and Saigusa 2004)
Controlled conditions (Hydroponic)	*Vitis vinifera* and interspecific varieties	2.00	(Blaich and Grundhöfer 1997)	Yes/NA	(Tyree 1994; Hermans et al. 2025)
Field	*Morus rubra* L	1.77	(Geis 1973; Hodson et al. 2005)	Yes	(Bozarth 1992)
Field	*Aronia melanocarpa* L	1.44	(Kolesnikov and Gins 2001; Hodson et al. 2005)	NA	
Field	*Theobroma cacao* L	1.29	(Klotz et al. 2023)	Yes	(Piperno 1985)
Field	*Corylus avellana* L	1.08	(Lisztes-Szabó et al. 2019)	Yes	(Lisztes-Szabó et al. 2019)
Field	*Vaccinium myrtillus* L	1.00	(Bartoli and Beaucire 1976; Hodson et al. 2005)	Yes	(Carnelli et al. 2002)
Controlled conditions and Field	*Corylus avellana* L	0.43—0.80	(Escobar-Hernández et al. 2025)	Yes	(Lisztes-Szabó et al. 2019)
Controlled conditions (Hydroponic)	*Coffea arabica* L	0.50	(Cunha et al. 2012)	Yes	(Astudillo 2019)
Field	*Vitis vinifera* L	0.40[b]	(Schabl et al. 2020)	Yes	(Tyree 1994; Hermans et al. 2025)
Controlled conditions (Pot)	Grapevine rootstocks: 1103P (*Vitis berlandieri* x *Vitis. rupestris*); 41 B (*Vitis vinifera* x *Vitis berlandieri*)	0.36—0.39	(Soylemezoglu et al. 2009)	NA	
Field	*Citrus sinensis* L	0.26[b]	(Mvondo-She and Marais 2019)	Yes	(Mvondo-She and Marais 2019)
Controlled conditions (Pot)	*Citrus reticulata* L	0.25[b]	(Mvondo-She and Marais 2019)	Yes	(Mvondo-She and Marais 2019)
Potted plants	*Vitis vinifera* L	0.21	(Noronha et al. 2020)	Yes	(Tyree 1994; Hermans et al. 2025)
Field	*Vitis vinifera* L	0.13	(Deshmukh et al. 2020)	Yes	(Tyree 1994; Hermans et al. 2025)
Controlled conditions (Pot)	*Olea europaea* L	0.025	(Nascimento-Silva et al. 2022)	Yes	(Tyree 1994; Tsartsidou et al. 2007)
Controlled conditions (Pot)	*Persea americana* Mill	0.008	(Dann and Le 2017)	Yes	(Sangster et al. 2001)
Field	*Malus pumila* Mill	0.003[b]	(Wang et al. 2016)	NA	
Field	*Castanea sativa* Mill	0.00013[b]	(Carneiro-Carvalho et al. 2019b)	Yes	(Carneiro-Carvalho et al. 2019a)
Agriculturally Important (timber and forest species)		Leaf Si ranges 0.3–11%; broad taxonomic coverage; phytoliths reported across multiple species and genera			

Table 1 (continued)

Growing Conditions	Woody species	Leaf Si (%[a])	References	Occurrence of phytoliths (Yes/No/Not Available (NA))	References
Field	*Fagus sylvatica* L	7.50—11.00	(Bartoli and Beaucire 1976; Hodson et al. 2005)	Yes	(Carnelli et al. 2002)
Field	*Abies alba* (ex *A. pectinata*)	5.30—6.25	(Bartoli and Beaucire 1976; Hodson et al. 2005)	Yes	(Lisztes-Szabó et al. 2019)
Field	*Pseudotsuga menziesii* Mirb	4.46	(Gérard et al. 2008)	Yes	(Lisztes-Szabó et al. 2019)
Field	*Celtis occidentalis* L	4.11	(Wilding and Drees 1971; Hodson et al. 2005)	Yes	(Lisztes-Szabó et al. 2019)
Field	*Broussonetia papyrifera* L	3.50	(Kajino and Kitajima 2021)	Yes	(Bozarth 1992)
Field	*Ulmus americana* L	2.38	(Lanning 1966; Hodson et al. 2005)	Yes	(Bozarth 1992)
Field	*Ulmus americana* L	1.54	(Geis 1973; Hodson et al. 2005)	Yes	(Bozarth 1992)
Field	*Quercus robur* L	1.42	(Lisztes-Szabó et al. 2019)	Yes	(Bozarth 1992)
Field	*Acer saccharum* Marsh	1.24	(Maguire et al. 2017)	Yes	(Bozarth 1992)
Field	*Acer saccharum* Marsh	1.23	(Geis 1973; Hodson et al. 2005)	Yes	(Radini et al. 2018)
Field	*Picea abies* L	1.20	(Hodson et al. 1997, 2005)	Yes	(Lisztes-Szabó et al. 2019)
Field	*Larix decidua* Mill	1.03	(Hodson et al. 1997, 2005)	Yes	(Pearsall 2016)
Field	*Picea orientalis* L	1.01	(Hodson et al. 1997, 2005)	Yes	(Lisztes-Szabó et al. 2019)
Field	*Fagus sylvatica* L	0.96	(Lisztes-Szabó et al. 2019)	Yes	(Lisztes-Szabó et al. 2019)
Field	*Picea abies* L	0.92	(Lisztes-Szabó et al. 2019)	Yes	(Lisztes-Szabó et al. 2019)
Field	*Abies alba* Mill	0.54	(Lisztes-Szabó et al. 2019)	Yes	(Lisztes-Szabó et al. 2019)
Field	*Pinus sylvestris* L	0.36	(Lisztes-Szabó et al. 2019)	Yes	(McCune and Pellatt 2013)
Field	*Pachira aquatica* Aubl	0.34	(Nakamura et al. 2024)	Yes	(Lisztes-Szabó et al. 2019)
Controlled conditions (peat)	*Betula pendula* Roth	0.14	(Hassan et al. 2025)	Yes	(Liu et al. 2021)
Field	*Picea* abies L	0.03	(Vartiainen et al. 2025)	Yes	(Lisztes-Szabó et al. 2019)
Additional tree and woody species (> 1% leaf Si and phytolith occurrence)			Leaf Si ranges 1–12.6%; phytolith occurrence aligns with higher Si concentrations		
Field	*Gironniera subaequalis* Planch	6.66	(Ishizawa et al. 2019)	Yes	(Kealhofer and Piperno 1998)
Controlled conditions (Pot)	*Phyllostachys pubescens* Mazel	3.70	(Liu et al. 2023a, b)	Yes	(Parr et al. 2010)

Table 1 (continued)

Growing Conditions	Woody species	Leaf Si (%[a])	References	Occurrence of phytoliths (Yes/No/Not Available (NA))	References
Field	*Ficus* sp.	1.30—2.70	(Nakamura et al. 2024)	Yes	(Dai et al. 2023)
Field	*Protium panamense* (Rose) I.M. Johnston	2.65	(Nakamura et al. 2024)	Yes	(Piperno 1989)
Field	*Calluna vulgaris* L	2.50	(Bartoli and Beaucire 1976; Hodson et al. 2005)	Yes	(Carnelli et al. 2002)
Field	*Sorocea affinis* Hemsl	2.29	(Klotz et al. 2023)	Yes	(Piperno and McMichael 2023)
Field	*Ligustrum lucidum* Ait	2.00	(Fernández Honaine et al. 2019)	Yes	(De Rito et al. 2018)
Field	*Duguetia* sp.	1.88	(Nakamura et al. 2024)	Yes	(Pearsall 2023)
Field	*Celtis occidentalis* L	1.61	(Geis 1973; Hodson et al. 2005)	Yes	(Bozarth 1992)
Field	*Talauma* sp.	1.49	(Nakamura et al. 2024)	Yes	(Piperno 1998)
Field	*Washingtonia filifera* Linden	1.43	(Lanning 1966; Hodson et al. 2005)	Yes	(McNamee 2013)
Field	*Ulmus macrocarpa* Hance	1.36	(Yang et al. 2015)	Yes	(Gao et al. 2018)
Field	*Herrania purpurea* Pittier	1.22	(Klotz et al. 2023)	Yes	(Piperno and McMichael 2023)
Controlled conditions (Pot)	*Populus trichocarpa* Torr. & Gray	1.20	(Deshmukh et al. 2020)	Yes	(Herndon et al. 2025)

[a]When necessary, original data was converted to Si%; [b]approximate median values from figure graphs

on natural soil processes (Etesami and Jeong 2020; Thakral et al. 2024; Rachappanavar et al. 2024).

These findings show that Si accumulation in fruit trees and woody species spans a wide spectrum, from low to high accumulators, with clear implications for both crop productivity and ecosystem functioning.

Phytolith occurrence

Numerous woody species belonging to diverse plant families (Tables 1–2) accumulate phytoliths and phytoliths have been recorded in economically significant fruit tree species (Table 1; Box 1).

Some species with measurable Si accumulation (Table 1) are also confirmed phytoliths producers such as *Vaccinium corymbosum* (Morikawa and Saigusa 2004), *Fagus sylvatica* (Tsartsidou et al. 2007), *Corylus avellana* (Lisztes-Szabó et al. 2019), and *Vitis vinifera* (Tyree 1994; Hermans et al. 2025).

Tree species can accumulate phytoliths (Cornelis et al. 2010b; Ge et al. 2020; An and Xie 2022) in leaves, other plant organs (Albert and Weiner 2001; Tsartsidou et al. 2007), and in different tissues such as bark (An and Xie 2022).

The highest leaf Si concentrations generally reported in tropical ecosystems (Ishizawa et al. 2019) highlight the possible influence of regional pedoclimatic conditions on phytolith dynamics. Compared to temperate forests, tropical forests are characterized by weathered acidic soils, intense leaching, high temperatures (Fujii et al. 2018) and display high biomass production and turnover (Schaller et al. 2018). Surveys show that a majority of tropical plant families are classified as abundant or common phytolith

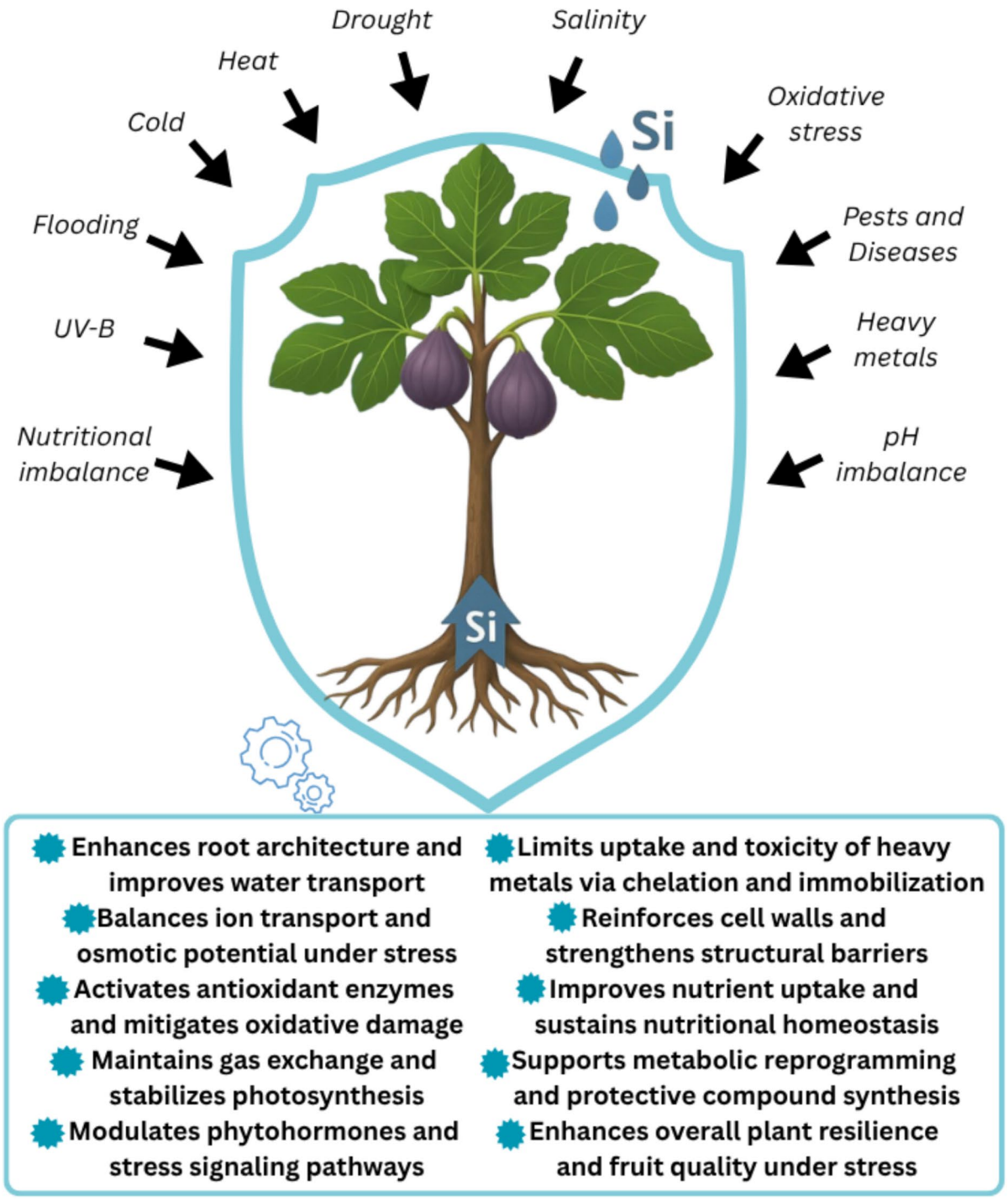


Fig. 1 Conceptual representation of stress tolerance in fruit trees associated with Si supplementation. The plant image was AI-generated (GPT®) and the figure was designed in Canva®. Silicon benefits are supported by literature (Etesami and Jeong 2020; Muhammad et al. 2022; Rachappanavar et al. 2024; de Souza Junior et al. 2025)

Table 2 Phytolith accumulation in woody species across various plant families

Families	Reference
Fagaceae, Magnoliidae, Pinaceae, Ulmaceae	An and Xie 2022
Adoxaceae, Anacardiaceae, Apocynaceae, Aquifoliaceae, Asteraceae, Boraginaceae, Brassicaceae, Celastraceae, Cistaceae, Cneoraceae, Convolvulaceae, Cupressaceae, Ericaceae, Fabaceae, Frankeniaceae, Globulariaceae, Hypericaceae, Lamiaceae, Lauraceae, Malvaceae, Moraceae, Myricaceae, Oleaceae, Pinaceae, Rhamnaceae, Rosaceae, Rubiaceae, Salicaceae, Sapotaceae, Scrophulariaceae, Tamaricaceae, Urticaceae	Castilla-Beltrán et al. 2024
Bignoniaceae, Burseraceae, Cannabaceae, Capparaceae, Caesalpinoideae, Chrysobalanaceae, Dipterocarpaceae, Euphorbiaceae, Fabaceae, Irvingiaceae, Malvaceae, Moraceae, Phyllanthaceae, Sapotaceae, Urticaceae	Collura and Neumann 2017
Anacardiaceae, Annonaceae, Apocynaceae, Calophyllaceae, Cannabaceae, Elaeocarpaceae, Erythroxylaceae, Euphorbiaceae, Lauraceae, Malvaceae, Meliaceae, Moraceae, Myrtaceae, Rubiaceae, Salicaceae, Sapindaceae, Sapotaceae, Urticaceae	de Oliveira et al. 2024
Adoxaceae, Moraceae, Rosaceae, Vitaceae	Hermans et al. 2025
Betulaceae, Fagaceae, Pinaceae	Lisztes-Szabó et al. 2019
Ericaceae	Morikawa and Saigusa 2004
Asteraceae, Cucurbitaceae, Fabaceae, Ulmaceae	Mulholland and Rapp 1992
Rutaceae	Mvondo-She and Marais 2019
Asteraceae, Cucurbitaceae, Fabaceae, Lauraceae, Ulmaceae	Sangster et al. 2001
Anacardiaceae, Betulaceae, Cannabaceae, Cupressaceae, Ericaceae, Fagaceae, Juglandaceae, Lamiaceae, Malvaceae, Moraceae, Oleaceae, Rosaceae, Salicaceae, Styracaceae, Tamaricaceae, Vitaceae	Tsartsidou et al. 2007
Oleaceae, Vitaceae	Tyree 1994

producers (Piperno 1988; Schaller et al. 2018). Yet, their soils typically contain less amorphous Si than temperate forests (Cornelis et al. 2010a; Schaller et al. 2018). Phytolith dissolution is faster at alkaline pH and inhibited under acidic conditions (Fraysse et al. 2009; Lu et al. 2022). In tropical forests, under understory N addition, Lu et al. (2022) reported that decreased soil pH reduced litter decomposition and limited the release of phytoliths from litter into topsoil. These results exemplify how regional soil properties play a role in shaping phytolith preservation and Si cycling.

Phytolith production in leaves is consistently observed across a broad spectrum of Mediterranean fruit trees and woody species present in the Mediterranean (Box 1). Collectively, these studies confirm consistent phytolith production in keystone Mediterranean fruit trees and woody perennials. This biochemical capacity underscores their value in Mediterranean agroforestry systems, where targeted integration can enhance Si cycling and contribute to ecological resilience.

Box 1. Leaf phytoliths have been recorded in *Ficus carica*, *Olea europaea*, and *Vitis vinifera* (Tyree 1994; Tsartsidou et al. 2007; Hermans et al. 2025), *Morus nigra*, *Prunus avium*, *Prunus domestica*, *Prunus spinosa*, *Pyrus communis*, *Rubus idaeus*, and *Sambucus nigra* (Hermans et al. 2025). Other species include *Quercus pubescens*, *Populus alba*, *Fagus sylvatica*, *Tilia* sp., *Acer monospessulanum*, *Celtis australis*, *Quercus frainetto*, *Carpinus betulus*, *Juglans regia*, *Quercus petraea*, *Alnus glutinosa*, *Pyrus amygdaliformis*, *Tamarix* sp., *Paliurus spina-christi*, *Crataegus* sp., *Phillyrea latifolia*, *Vitis vinifera sylvestris*, *Pistacia terebinthus*, *Quercus coccifera*, *Arbutus andrachne*, *Vitex agnus-castus*, *Prunus amygdalus*, *Fraxinus ornus*, *Juniperus oxycedrus*, and *Sorbus torminalis* (Tsartsidou et al. 2007), and *Erica arborea* (Castilla-Beltrán et al. 2024). Reproductive tissue phytoliths occur in *Ficus carica*, *Morus nigra*, *Vitis vinifera*, *Prunus cerasus*, and *Rubus* spp. (Hermans et al. 2025).

Certain dicot families (Acanthaceae, Cannabaceae, Moraceae, and Urticaceae) accumulate silica in cystoliths, calcium carbonate bodies with silicified stalks connected to cell walls that contribute to elevated leaf silica content (Fernández Honaine et al. 2023). Silicon also accumulates in trichomes (Piperno 2006), which are known to enhance plant resilience

by protecting against stresses (Karabourniotis et al. 2020). In some families (i.e., Cannabaceae, Boraginaceae, and Loasaceae), silica and calcium deposits localize at the bases of trichomes or adjacent cells (Fernández Honaine et al. 2023). Trichomes appear to be silica-accumulating structures that may complement phytolith formation in dicots, and warrant further investigation. In glandular trichomes, it has been recently reported a lignin-based cell wall structure named 'neck strip', similar to the Casparian band present at root level, achieving compartmentalization by acting as an extracellular barrier, a condition crucial for silica polymerization (Hao et al. 2024).

Phytolith production has also been confirmed in the woody and bark tissues of *Ficus carica, Morus nigra*, and *Rubus* spp. (Hermans et al. 2025), and other relevant Mediterranean species (Albert and Weiner 2001; Tsartsidou et al. 2007; Castilla-Beltrán et al. 2024). Silica is also frequently found in wood (Scurfield et al. 1974), used for wood identification and classification (Siam et al. 2022), and which can influence wood density and quality (Romero et al. 2020).

In summary, the capacity of woody species to absorb, translocate, and deposit Si, return it to the soil through durable phytoliths, positions these species as central actors in nutrient cycling and stress mitigation. Incorporating Si-accumulating woody species (Table 1) into AF systems can contribute to ecosystem functioning (Song et al. 2014; Katz et al. 2021). Considering Si's potential role in shaping species performance and ecosystem responses to climate change in certain ecosystems (Schaller et al. 2018), studying resilience mechanisms in ecosystems whose flora has adapted to extreme environmental conditions (Matesanz and Valladares 2014), can provide further insights for agricultural development and ecosystem restoration.

Si-based agroecological and agroforestry models

As the need for resilient and sustainable agroecosystems grows, the valorization of key strategies to enhance ecosystem services and adapt to climate change has gained recognition. Agroecology is based on mixed farming models that prioritize biodiversity and ecological principals, with AF encompassing holistic-farming systems that integrate woody species, herbaceous crops and/or livestock (Rombolà 2022; Quandt et al. 2023; Nungula et al. 2024).

Althought limited, research on Si within AE and AF systems indicates potential advantages in diversified cropping systems. Evidence shows Si-rich livestock manure and AF systems increased Si content in banana (Fujii et al. 2023). Intercropping rice and water spinach enhanced rice yield, pest control, and Si nutrition through rhizosphere interactions and increased Si availability (Ning et al. 2017; 2022). In sugarcane, intercropping with peanut, coupled with Si application, improved soil nutrition and sugarcane yield (Shen et al. 2019), and Si supplementation in a legume agroecosystem influenced crop productivity and arthropod communities (Putra et al. 2024). Rice-soybean intercropping with Si application enhanced yield production through root morphology modification and distribution (Zhang et al. 2025). Likewise, Si-enriched compost combined with beneficial microbes improved rice performance in AE systems (Llerena-Ramos et al. 2025). These results highlight how Si bioavailability is influenced by rhizosphere mechanisms in diversified systems, including microbial activity, root exudation, and soil processes.

Crop diversity plays a key role in enhancing nutrient availability and ecosystem functionality (Yu et al. 2022). Intercropping has been shown to facilitate acquisition of certain nutrients through below-ground interactions (Xue et al. 2016; Yu et al. 2022), this sustainable approach has been investigated and applied in fruit tree cultivation (Tagliavini and Rombolà 2001; Cesco et al. 2006).

Integrating AE and AF system principles and Si ecological interactions offers an innovative response to regions such as the Mediterranean. Despite their known biodiversity and agricultural value (Matesanz and Valladares 2014), these regions face threats like climate change, soil erosion, land abandonment and agricultural intensification (Matesanz and Valladares 2014). Although the adaptive traits of Mediterranean woody species have been extensively studied (Zunzunegui et al. 2011; Medda et al. 2022), Si-related processes are limited (de Tombeur et al. 2020; Moura et al. 2024). Recent studies revealed that Si processes in Mediterranean and arid ecosystems are influenced by carbonate and silicate weathering domains with implications in long-term soil development (de Tombeur et al. 2020) and that Si uptake influences plant community assembly and ecosystem structure (Moura et al. 2024).

The introduction of trees with herbaceous cultivation has come into the spotlight as the concept of

agroforestry has shown multiple benefits for environmental and economic sustainability (Lehmann et al. 2020; Nungula et al. 2024). Several woody species can accumulate significant concentrations of Si (Table 2) and Si has proven beneficial in fruit and woody species (Ladon et al. 2021; Muhammad et al. 2022; Rachappanavar et al. 2024). Preliminary work in species present in the Mediterranean maquis *Quercus ilex*, *Fraxinus ornus*, *Arbutus unedo, Erica arborea* and *Olea europaea* showed species-specific and site-dependent Si accumulation in leaves (Rodríguez Declet 2025). Phytolith production is documented in *Olea europaea* (Tyree 1994), *Fraxinus ornus* (Castilla-Beltrán et al. 2024) and *Erica arborea* (Castilla-Beltrán et al. 2024). Other key Mediterranean maquis species (*Quercus ilex*, *Arbutus unedo*) remain understudied, despite known phytolith production in related species (Box 1; Tsartsidou et al. 2007), although *Q. ilex* may produce few diagnostic morphotypes suitable for reliable identification (Bremond et al. 2004). Given phytoliths' (Box 1) dual role in Si cycling and C sequestration, quantifying their production in these species could prove valuable for Mediterranean ecosystem regeneration.

Si-accumulator species can enhance Si availability by promoting mineral dissolution (Harley and Gilkes 2000; Sugiyama and Ae 2000; Akagi et al. 2006), which improves plant growth in weathered or nutrient-depleted soils. Additionally, microbes isolated from silicate minerals (Sheng et al. 2008; Wang et al. 2015), plant-associated bacteria (Raturi et al. 2021), and mycorrhizal fungi (van Schöll et al. 2008; Quirk et al. 2012) mobilize Si from soil minerals (Etesami et al. 2021; Kothari et al. 2025). Plants facilitate mineral weathering and Si bioavailability through the production of weathering agents in the rhizosphere (Bennett et al. 1988; Kelly et al. 1998; Hinsinger et al. 2001). Root exudates, such as the carboxylates citric acid, malic acid, and oxalic acid, enhance the solubilization of nutrients like phosphorus (P) and manganese (Mn) and have been shown to also mobilize Si from minerals (de Tombeur et al. 2021), leading to elevated leaf Si concentrations (Yan et al. 2002; Lambers et al. 2015; de Tombeur et al. 2021). Nitrogen (N) addition in subtropical forests has been linked to increased carboxylate exudation, also enhancing silicate weathering, Si availability and nutrient mobilization (Yu et al. 2024). Recently Kostic et al. (2025), as a consequence of adding ammonium (NH_4^+), reported a decrease in rhizosphere-soil pH associated to an increase in the rhizosphere-soil mobile Si fraction which was attributed to diminished Si adsorbed to pedogenic Fe/Mn oxides and Si phytogenic fractions. Though nutrient acquisition strategies based on root exudation are considered energetically expensive (Raven et al. 2018), as highlighted by de Tombeur et al. (2023), the costs of carboxylates and other Si-mobilizing root exudates may be negligible if they have already been covered by other processes (e.g., to mobilize soil P) or if they result from a surplus of C during nutrient or water deficiencies (Prescott 2022) or stressed conditions (Canarini et al. 2019; Prescott 2022). Silicon has also been shown to improve the acquisition and utilization of essential elements in plants (Pavlovic et al. 2021) enhancing nutrient availability in the rhizosphere, by interacting with well-known soil processes.

Si-accumulating species (Epstein 2009), sugarcane and *Urochloa* spp. may influence mineral weathering through rhizosphere acidification resulting from root proton excretion, exudation of organic acids, and increased CO_2 production associated with heightened biological activity, promoting the dissolution of primary minerals like quartz and releasing Si and essential nutrients (Ruiz et al. 2022). Following senescence, Si contained in plant residues returns to the soil favoring the stability of clay minerals, influencing pedogenesis (Ruiz et al. 2022).

Phytosiderophores released by grasses (Römheld and Marschner 1986) and sugarcane (Figueira et al. 2001) may interact with Si increasing its availability presumably through ligand-promoted dissolution. Gattullo et al. (2016) reported that the dissolution of smectite induced by barley was regulated by organic ligands exuded by roots, among which phytosiderophores and organic acids. Noteworthy, phytosiderophes have also been found in olive trees (Suzuki et al. 2016).

The rhizosphere exerts a strong influence on phytolith dynamics. As previously discussed, root exudates mobilize Si from mineral and phytogenic pools, including phytoliths, particularly under nutrient-limited conditions (de Tombeur et al. 2021 and references therein). Recently, Wei et al. (2025) reported that organic acids exudation and Si uptake increased in response to bicarbonate, an anion known for its marked effects on organic acids metabolism (Abadía et al. 2002; Covarrubias and Rombolà 2013). Root exudation influences rhizosphere microorganisms (Huang et al. 2014), including microorganisms involved in silicate weathering (Etesami et al. 2021;

Raturi et al. 2021) and potentially in phytolith dissolution. Rhizosphere and soil pH impact phytolith dissolution (Fraysse et al. 2009; Kostic et al. 2025). Soil aggregates in the rhizosphere can physically protect phytoliths from dissolution, extending their persistence and stabilizing the biogenic silica pool (Li et al. 2022; de Tombeur et al. 2024).

Perennial and woody species can produce deep roots extending from 2.5 to > 53 m (Smart et al. 2006; Ollat et al. 2015; Pierret et al. 2016) which can enhance mineral weathering through rhizosphere and water uptake processes (Thorup-Kristensen et al. 2020), increase C-sequestration and improve plant water-use efficiency (Peixoto et al. 2022). Trees enhance the release of dissolved Si from soil minerals (Cornelis et al. 2010a), and transport Si to the topsoil from deep, less weathered soil layers (Cornelis and Delvaux 2016), as seen in tropical forests, where deep rooting systems of evergreen trees facilitate the upward transfer from lithogenic and pedogenic silicates (Alexandre et al. 1997; Cornelis and Delvaux 2016). Developing deep-rooted crops varieties may similarly enhance Si uptake and agroecosystem services (Li et al. 2025).

Research highlights the significant role of tree species and tree organs in biogeochemical Si cycles and therefore ecological roles (Cornelis et al. 2010b; Schaller et al. 2018; Nakamura et al. 2019). The Si cycle is closely linked to C sequestration in agricultural ecosystems, as Si-related C sinks tend to be more stable than other sinks (Song et al. 2014). The recycling of Si by forest trees contributes significantly to the continental Si cycle (Cornelis et al. 2010b) and has implications for global biogeochemical processes (Cornelis et al. 2010b; Kothari et al. 2025).

Phytolith rich biomass serves as a vital sink of soluble Si in agricultural and forest ecosystems (Bartoli 1983; Alexandre et al. 1997; Gérard et al. 2008; Cornelis et al. 2010b). In forests, Si is largely present as biogenic Si, primarily from phytoliths and adsorbed Si (Cornelis et al. 2011). Biogenic Si replenishes dissolved Si pools, outperforming silicate weathering in soil Si release (Alexandre et al. 1997; Rezakhani et al. 2022) and also contributes to pedogenesis by fostering the formation of new mineral structures in specific environments (Lucas 2001; Wüst and Bustin 2003; Borrelli et al. 2008). Thus, these sources play a significant role in long-term soil development and Si fluxes, influencing soil fertility and ecosystem dynamics (de Tombeur et al. 2021; Cornu et al. 2022; Li et al. 2022). Amorphous Si (ASi), which includes biogenic ASi (e.g., phytoliths, protist shells, sponge spicules) and minerogenic ASi (e.g., silica nodules, silica included in pedogenic oxides/hydroxides), together with dissolved silicic acid, constitutes reactive silica, a chemically highly reactive fraction of silica, whose depletion has been identified as an early indicator of soil degradation, often preceding declines in soil structure, microbial activity, and overall soil fertility (Schaller et al. 2025).

After litterfall or root turnover, phytoliths enter a dynamic soil-Si-C feedback system. Their persistence and solubility are influenced by pH, soil organic matter dynamics, Fe-oxide adsorption, water availability, and aggregation status (Cornelis and Delvaux 2016; Li et al. 2022; de Tombeur et al. 2024). Organic matrices and aggregate structures can reduce dissolution rates, enhancing the long-term stability of both silica and occluded C (Fraysse et al. 2009; Li et al. 2022; de Tombeur et al. 2024). Phytolith chemistry, and by extension their stability in soil, varies by plant type, ecological composition and structure (Song et al. 2017). Tree species with specific phytolith morphologies or chemical compositions may resist decomposition or dissolution more effectively, altering the residence time of biogenic Si in soil (Blecker et al. 2006; Song et al. 2017). Soil management strategies of agricultural soils should consider these interactions to improve the soil C and plant-available Si reservoir.

Phytoliths improve soil fertility by contributing to soil aggregation, enhancing porosity and water retention; additionally, their gradual dissolution supplies reactive silica, supports microbial habitats and mediate rhizosphere processes (Schaller et al. 2021; de Tombeur et al. 2024). During silica precipitation, small fractions of organic compounds become physically encapsulated, forming PhytOC (Hodson 2019; de Tombeur et al. 2024). Carbon interacts with phytoliths through organic compounds and mineral elements accumulated in the cell wall and lumen (Hodson 2019; de Tombeur et al. 2024). High-resolution synchrotron analyses demonstrate that this C occurs as lignin-like and proteinaceous compounds located at phytolith borders, within silica islands, and as mixed silica-C matrices, highlighting how it is physically protected within the silica (Negrão et al. 2024). Similarly, nanoscale SEM–EDX and STEM–EDX

analyses showed that C and Si are co-localized within cell-wall phytoliths, with heterogeneous distributions across different structural domains (de Tombeur et al. 2024). Organic C occluded within the silica matrix exhibits high persistence in soils (Cheng et al. 2023), as it protects C from microbial decomposition and oxidation, allowing PhytOC to persist (Song et al. 2017; Zhao et al. 2025). However, stability depends on phytolith type and soil conditions, as dissolution can release associated organic matter and stimulate soil organic carbon mineralization (de Tombeur et al. 2024). PhytOC is recognized as a stable C pool which highlights the need for targeted research (de Tombeur et al. 2024).

Understanding plant diversity, soil processes, and phytolith deposition is essential for leveraging Si as a soil-building and ecosystem-regulating element in diversified agricultural systems. From an agroecological perspective, fruit trees and other woody species represent under-explored vectors for Si cycling and PhytOC formation, especially in Mediterranean AF landscapes where biodiversity and stress-tolerance modulate biogeochemical processes. Bridging gaps in Si accumulation across tree organs and methods to trace PhytOC in soils are essential for linking Si dynamics into terrestrial C models and climate mitigation. Silicon return via phytolith-rich litter from trees can stabilize soil chemistry, support C sequestration, and reinforce ecosystem services, as Si also enhances the resilience to biotic and abiotic stresses of multiple tree species (Kothari et al. 2025). Integrating deciduous species with high Si cycling rates can optimize nutrient recycling (Kothari et al. 2025), reducing dependence on synthetic inputs and improving overall system productivity. These ecological services alongside diversified crop rotation and intercropping in AE and AF systems can enhance Si cycling by incorporating species with complementary Si uptake and recycling rates (Fig. 2).

Many herbaceous species with proficient Si accumulation (Ma and Yamaji 2006; Katz et al. 2021) coupled with the deep-rooting capacity of trees to increase mineral elements accessibility and availability (Callesen et al. 2016; Thorup-Kristensen et al. 2020) and improve water access (Thorup-Kristensen et al. 2020), further drive Si cycling and strengthen the sustainability and resilience of agroecosystems. Additionally, enhanced rhizosphere activity accelerates mineral weathering, liberating Si and other essential nutrients (Harley and Gilkes 2000; Sugiyama and Ae 2000; Akagi et al. 2006; de Tombeur et al. 2021), thereby optimizing nutrient cycling, reducing dependence on external inputs, and improving soil fertility in AE and AF systems.

Incorporating Si-accumulating woody species into AF systems can contribute to soil fertility through multiple pathways. They may act as sources of phytoliths (Cornelis et al. 2010b, 2011), replenishing soil ASi pools, increasing soil fertility and health indices such as nutrient availability, C fixation, aggregate stability, water availability and storage, and microbial abundance and community (Schaller et al. 2025). Woody species exhibit contrasting phytolith accumulation (Castilla-Beltrán et al. 2024; Hermans et al. 2025), which could guide species choice in AF systems alongside their agronomic, environmental and economic benefits. Although woody species generally produce relatively low number of leaf phytoliths (Zhao et al. 2025), they can contribute to Si cycling through Si deposition in other organs (Hodson 2019). Moreover, woody species, both low and high phytolith accumulators, can influence soil Si availability through the accumulation of non-phytolithic Si forms and/or processes other than Si root uptake (e.g., rhizosphere pH changes).

AE and AF systems can exploit Si-mediated processes and cycling, with some potential examples including: Si recycling by leaf and wood litter, promotion of Si-accumulation by Si efficient rootstocks, Si pumping by deep-rooted tree species, microorganisms and mycorrhizal interactions, promotion of mineral weathering by Si-accumulating plants and microorganisms, and PhytOC sequestration (Fig. 2).

For example, in viticulture, a traditionally important socio-economic sector in the Mediterranean region, rootstocks became crucial for managing abiotic and biotic stress, however there are research gaps in the knowledge and selection of rootstocks for addressing emerging challenges (Chen et al. 2024). Considering the known role of Si in overcoming stress related impacts and grapevine's Si uptake genes (Noronha et al. 2020), current approaches still do not specifically discuss the physiology of Si nutrition in relation to rootstocks and grapevine scion (Chen et al. 2024). In this context, experiments conducted on grapevine grafted with different rootstocks indicated that certain rootstocks induced higher Si accumulation (Rodríguez Declet 2025). Grapevine can develop deep

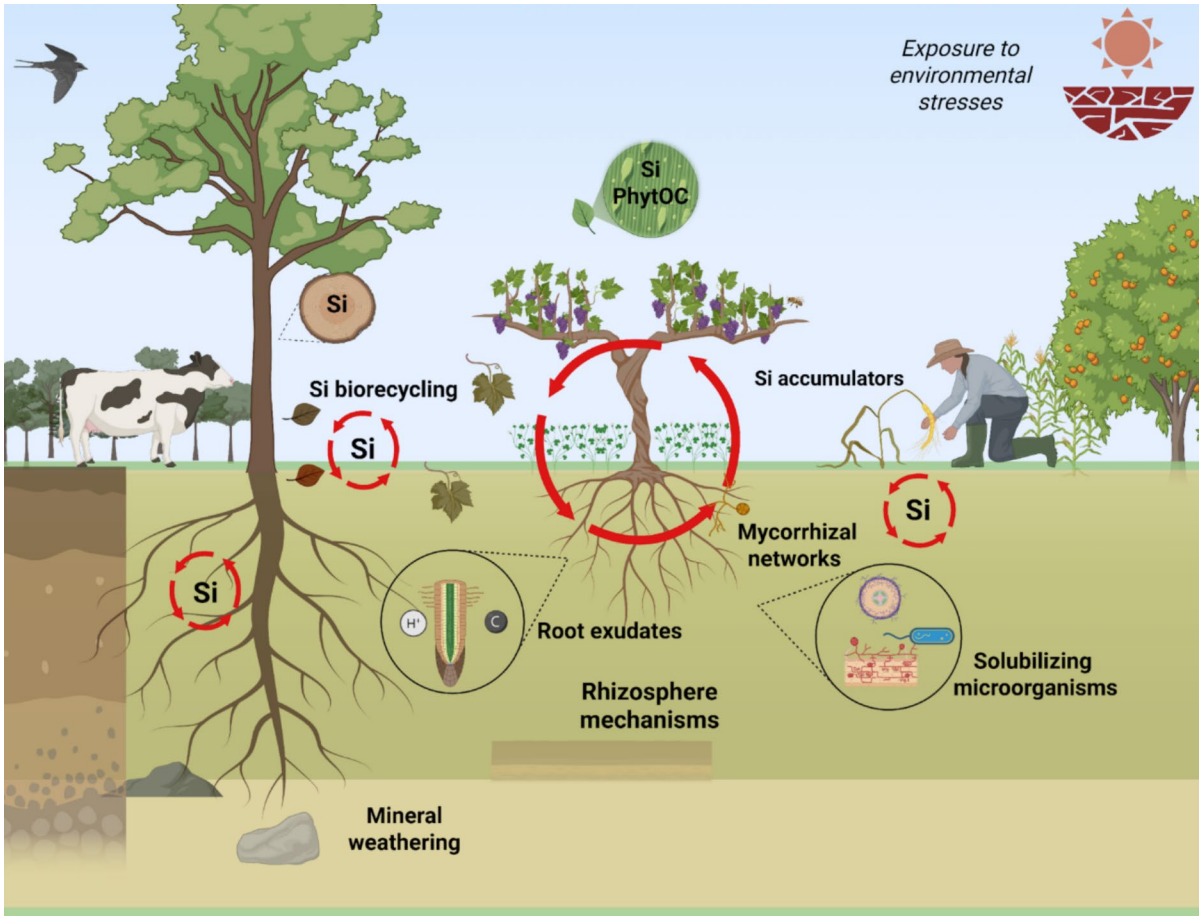


Fig. 2 Schematic representation of AE and AF model system based on Si processes. Trees with deep-root systems access subsoil Si pools, driving mineral weathering, accumulating and returning Si and phytoliths to the superficial soil layers, contributing to Si cycling. Harnessing Si and other nutrient cycling in AE and AF systems through Si-accumulating species, rhizosphere interactions, microorganisms, mycorrhizal networking, is functional for agroecosystem resilience. Created with BioRender.com

root systems (Ollat et al. 2015), which may promote Si soil processes (Cornelis and Delvaux 2016) and more efficient use of natural resources (Peixoto et al. 2022). Other relevant Mediterranean species, such as olive, may provide valuable insights into processes of resource acquisition and allocation in vulnerable conditions. Understanding the differences in Si and PhytOC accumulation between and within species is also highly relevant (Hodson 2019), for their ecosystem functions in the agroecosystems.

Furthermore, the re-valorization of traditional and improved AF systems (Favor and Udawatta 2021) and AE and AF management in viticulture (Rombolà 2022) and other long-established Mediterranean systems (i.e., olive, citrus, fig, almond, hazelnut, chestnut, walnut) (Eichhorn et al. 2006; Ramachandran et al. 2017) have surfaced as highly valuable strategies for their known resilience to extreme climate events (Quandt et al. 2023; Nungula et al. 2024) and multiple ecosystems services (Nungula et al. 2024). This further promotes the development and implementation of model fruit-tree AE and AF systems (Barbault et al. 2024). Under controlled environment conditions, intercropping grapevine with *Fraxinus ornus* (manna ash), enhanced Si concentration in grapevine leaves (Rodríguez Declet 2025). These findings emphasize the potential benefits

of Si-based strategies and systems for enhancing plant resilience and promoting their integration into agroforestry systems.

The relevance attributed to Si accumulators is supported by evidence demonstrating that Si and elements released from primary minerals (Hinsinger et al. 2001; Akagi et al. 2006; de Tombeur et al. 2021; Ruiz et al. 2022) and from phytoliths (Farmer et al. 2005; Schaller et al. 2021) can contribute to plant nutrition. These may have implications on fertility indices (e.g., exchangeable and assimilable fractions of mineral elements), largely used by soil chemists and agronomists as indicators of nutrient availability for crops.

In papers discussing the effects of Si-accumulators in agricultural systems emphasis has often been given to soil desilication phenomena which these species can cause (Haynes 2014), rather than to their capability of activating bidirectional Si fluxes in the plant-soil system, a typical feature of forest systems (Haynes 2014).

The targeted integration of Si-accumulating trees, combined with the valorization of traditional Mediterranean systems and intercropping soil management strategies offers practical pathways for building resilient agroforestry systems.

Conclusions

Fruit trees and other woody species exhibit measurable Si accumulation across diverse taxa and environmental conditions, and Si application in fruit trees and woody forest species are beneficial for productive capacity and physiological performance. Several fruit trees and other woody species reach foliar Si concentrations above 1%, and phytolith production is consistently observed in plant tissues and organs, positioning them as relevant contributors to Si cycling, with implications for nutrient availability, soil structural integrity, PhytOC formation, and overall ecosystem restoration processes.

Existing knowledge on Si cycling in forest ecosystems and rhizosphere processes can be applied in AE and AF systems. Silicon accumulation by deep-rooted woody species may lead to increased availability of Si and other elements in the soil by mineral weathering and Si-recycling by leaf and wood litter and root turnover. These mechanisms, combined with appropriate soil management practices, can improve Si nutrition and resource efficiency in intercropped systems. The adoption of agronomic strategies, such as Si-efficient rootstocks and intercropping with Si-accumulating species, could offer new pathways for improving agroecosystem resilience.

The ability of some woody species to maintain high Si cycling in the soil–plant system, solubilize and take up Si, produce, accumulate and return phytoliths, is configured as a vital ecophysiological and agronomic trait for counteracting climate change, and may also contribute to safeguard and improve forest performance and management.

Advancing AE and AF management strategies and system designs that consider Si dynamics could address key challenges faced by vulnerable agricultural regions, such as the Mediterranean, supporting nutrient cycling, enhancing C sequestration, improving soil quality, and reducing or avoiding the dependence on synthetic inputs.

Priority areas for future research include:

1. Molecular and physiological characterization of Si transporters in fruit trees and other woody species.
2. Extending phytolith quantification and classification of economically relevant fruit trees and other woody species.
3. Modelling of Si fluxes within AE and AF system designs, with specific emphasis on Mediterranean and climate-vulnerable agroecosystems, to quantify Si inputs, cycling, reactive silica and feedback loops. Paying particular attention to field-based, long-terms experiments.
4. Conducting investigation of rootstock–scion interactions as modulators of Si uptake, particularly in perennial fruit systems reliant on grafting strategies.
5. Long-term assessment of phytolith returns on soil fertility, microbial community composition, and C stabilization dynamics.
6. Si-accumulating plants and Si contribution in modulating the interception and allocation of solar energy, transformed in chemical energy by the production of photosynthates, Si-compounds and energy-rich molecules flowing into the soil through the rhizosphere.

Recognizing the functional relevance of Si in perennial cropping systems, through transdisciplinary research will provide a foundation for advancing the design of AE and AF systems, climate-resilient agroecosystems that optimize biogeochemical cycling, enhance ecological performance, and support long-term agricultural sustainability.

Acknowledgements We sincerely appreciate the anonymous reviewers for their valuable time and insightful comments, which have significantly enhanced the quality of our manuscript.

Author contributions This review is part of the PhD thesis of Arleen Rodríguez Declet.

Funding Open access funding provided by Alma Mater Studiorum - Università di Bologna within the CRUI-CARE Agreement. The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Data availability All data generated or analyzed during this study are included in this published article.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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